American Biochar Institute

Biochar Effects on Soil Water Retention and Water Movement

Scientific Basis for Improving Irrigation Efficiency

Irrigation accounts for the largest share of consumptive freshwater use in U.S. agriculture, and rising water costs, declining aquifers, and more frequent drought are sharpening the need for practices that let crops do more with less applied water. Soil amendments that increase the volume of water a soil can hold and supply to roots between irrigation events directly target this problem. Biochar, the carbon-rich solid produced by pyrolyzing biomass under low-oxygen conditions, has been studied extensively for this purpose.

This document summarizes the peer-reviewed evidence on how biochar changes two linked soil properties central to irrigation efficiency: **water retention** (how much water the soil holds and how much of it plants can use) and **water movement** (how quickly water infiltrates, redistributes, and drains). It emphasizes where the effects are strong, where they are weak or absent, and what governs the difference, so that the science can be matched honestly to field decisions.

1. Key Concepts

Three core measures frame how biochar affects irrigation-relevant soil water:

- **Field capacity (FC):** the water a soil retains after gravity drainage
- **Permanent wilting point (PWP):** the water held so tightly that roots can no longer extract it.
- **Plant-available water (PAW):** the difference between FC and PWP, also called available water capacity (AWC). This is the usable reservoir that buffers crops between irrigation events.

For irrigation efficiency, PAW is pivotal. Increasing it allows a field to store more usable water per unit depth of soil, which can lengthen the interval between irrigations, reduce deep percolation losses, and lower pumping volumes.

Water movement is described primarily by **saturated hydraulic conductivity (Ksat)**: the rate at which water moves through saturated soil. Ksat influences how:

- Quickly irrigation water infiltrates the soil;
- Much water runs off the surface;
- Rapidly water drains below the root zone.

The key point is that retention and movement are linked. The same pore-structure changes that increase water storage can also change how quickly water enters, redistributes, and leaves the soil. Depending on soil texture, these effects may reinforce each other or partially offset one another.

2. Evidence

Several independent meta-analyses show similar results: **biochar generally increases soil water retention and plant-available water, but the magnitude is highly conditional.**

In a literature synthesis through 2015, Omondi et al. found that biochar increased available water holding capacity by about 15% and saturated hydraulic conductivity by about 25%. It also lowered bulk density by roughly 8% and raised porosity by about 8%.[3]

Edeh et al. examined a broader set of hydraulic parameters and reported average increases in available water content of about 29%, field capacity of about 20%, and total porosity of about 9%, alongside a roughly 39% decrease in saturated hydraulic conductivity.[2] **Importantly, the Ksat response depended on texture: biochar reduced Ksat in sandy soils but increased it in clayey soils, helping reduce runoff in the latter.[2]**

Razzaghi et al. most clearly quantified how retention benefits vary by texture; their findings are summarized below.[1]

Table 1. Average change in soil water parameters after biochar amendment, by soil texture (Razzaghi et al., 2020).

Soil texture	Field capacity	Wilting point	Plant-available water
Coarse (sandy)	+51%	+47%	+45%
Medium (loam)	+13%	+9%	+21%
Fine (clay)	<1% (n.s.)	−5%	+14%

Bulk density decreased by ~9% across all textures. n.s. = not statistically significant.

The pattern is similar across all studies: **the retention benefit is largest in coarse, sandy soils and smallest — sometimes negligible — in fine, clayey soils.**[1,2,4]

A texture-specific meta-analysis by Wei et al. likewise found field-capacity and available-water gains roughly two to three times larger in coarse soils than in fine soils.[4] For an irrigation-efficiency program, this is the single most important planning fact: **the soils that benefit most are exactly the droughty, low-retention sandy soils where irrigation demand is typically highest.**

3. Why Biochar Changes Soil Water Behavior

The hydrological effects of biochar arise from several interacting physical mechanisms rather than a single cause.[8,17]

Internal porosity and high surface area

Biochar particles retain the cellular pore structure of their feedstock, giving them very high internal porosity and specific surface area. These intra-particle pores store water directly and contribute additional fine pore space that holds water in the plant-available suction range.[8] Adhikari et al. identify pore architecture, surface area, and surface chemistry as the dominant controls on a biochar's intrinsic water-holding capacity.[8]

Particle size, shape, and pore reorganization

Biochar affects soil water not only through its own internal pores, but also by reorganizing the pore space in the soil around it. Particle size and shape are therefore important design factors.

- **Intrapores** are pores within biochar particles. They can increase water content when soil is relatively dry.
- **Interpores** are spaces between soil and biochar particles. Irregular, elongated biochar particles can disrupt sand-grain packing when soil is wet, creating additional storage space.

Liu et al. used controlled sand-matrix experiments to distinguish these two pore systems. Their practical conclusion was that biochars with **high intraporosity and irregular shapes are most effective at increasing water storage in coarse soils.**[6]

Particle size does not affect every soil the same way, and the way water content is reported can change the apparent size of the benefit.

- Zhang et al. tested hydrophilic biochar in sand, silt loam, and clay across four particle-size fractions. **Finer particles increased plant-available water in silt loam and clay but had little effect in sand.**[7]
- Because biochar lowers bulk density, gains reported on a gravimetric basis can look larger than gains reported on a volumetric basis. This helps explain why laboratory column results often appear stronger than field results.[7]
- Verheijen et al. similarly found that finer particles and higher application rates lowered bulk density and increased maximum water holding capacity in sandy and sandy loam soils, but the returns diminished and depended on texture.[13]

Table 2. Practical particle-size and rate guidance from the experimental and meta-analytic literature.

Soil texture	Biochar guidance for water benefit	Source basis
Coarse / sandy (>60% sand)	Small particle size (<2 mm), high surface area and intraporosity; irregular shapes; excessive application rates give the largest retention gains.	Edeh; Liu; Zhang
Medium / loam	Finer particles raise volumetric available water; gains moderate; verify against bulk-density change.	Zhang; Wei
Fine / clay (>50% clay)	Lower rates of high-surface-area biochar; benefit chiefly via improved Ksat and structure, not added storage.	Edeh; Wei

Gravimetric vs volumetric reporting affects apparent magnitude; field gains are typically smaller than laboratory column results.

Lower bulk density and higher total porosity

Biochar is light and porous, so adding it lowers soil bulk density and raises total porosity — effects seen consistently across meta-analyses.[2,3] Blanco-Canqui's synthesis found biochar reduced bulk density and can relieve compaction, indirectly improving both water storage and root access.[14]

4. Effects on Water Movement, Infiltration, and Drainage

Texture controls the direction of water-movement effects. Because biochar changes pore structure, it can alter saturated hydraulic conductivity (Ksat) in opposite ways depending on soil texture:

- **Sandy soils:** biochar tends to lower Ksat, slowing drainage and helping water remain longer in the root zone.
- **Clayey soils:** biochar tends to raise Ksat by opening structure, improving infiltration and reducing surface runoff, ponding, and related losses.[2]

Wang et al. showed that these movement changes are tied to shifts in the soil water retention curve, not simply to one bulk soil property.[5]

In coarse soils, slower drainage can reduce deep percolation, keeping more water and dissolved nutrients within the root zone. This can improve both water-use efficiency and nutrient retention, and field and column studies attribute reduced leaching partly to this retention effect.[2,3]

Need for site-specific verification. *The broader literature emphasizes that water-movement responses depend on the specific biochar–soil pairing and should be measured or modeled rather than assumed.[17] Consistent with that caution, Ren et al. found that higher biochar rates slightly increased simulated infiltration and reduced evaporation; field-capacity and saturation effects scaled with initial soil organic carbon, while wilting point changed little.[18]*

5. From Soil Water to Irrigation Savings

Models link improved retention to lower irrigation demand. Kroeger et al. combined biochar water-holding data with a county-level U.S. economic model and estimated about a 37% reduction in irrigation use at a sandy-soil site in Nebraska.[12] The largest water-related returns were projected for sandy soils in the western, far-northern, and southeastern U.S.[12]

Field results are less consistent. Aller et al. found that fresh and aged biochars increased retention in clay loam but reduced plant-available water and water-use efficiency; effects in sandy loam and silt loam were positive or mixed.[9]

The key lesson: more stored water helps only if it remains within the plant-available suction range.

6. Executive Review

The evidence base carries real caveats that a credible, funder-facing proposal should state plainly:

- **Texture dependence.** Benefits are large in sand and small in clay; fine-textured soils may see no field-capacity gain and even a slight wilting-point increase.[1]
- **Mixed results.** Jeffery et al. found biochar did not improve the hydrological function of a sandy soil, showing the coarse-soil benefit is not guaranteed.[10]
- **Application rate and economics.** Meta-analyses point to absurdly high application rates for water effects, quantities that carry significant material and application cost relative to the per-acre water savings.[2]
- **Feedstock and pyrolysis variability.** Particle size, surface area, and porosity all depend on feedstock and production temperature, so results from one biochar do not transfer cleanly to another.[6,8]
- **Aging and time scale.** Benefits may emerge or shift as biochar weathers, and *long-term field data remain comparatively scarce*, though a 10-year vineyard study by Baronti et al. documented persistent improvements in soil-plant water relations.[16]
- **Field-scale prediction is still maturing.** A 2026 global model evaluation by Ren et al. reproduced biochar effects well in temperate and tropical systems but performed worst in arid zones and on coarse soils — precisely the dry, sandy settings most relevant to irrigation — and the authors flag biochar–soil-moisture interaction and irrigation scheduling under drought as unresolved modeling priorities.[18]

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